

Overview of Chain of thought reasoning

Subject: Chain of thought reasoning

1.

Automated prompt generation Recent research has explored automated prompt engineering, using optimization algorithms to generate or refine prompts without human intervention. These automated approaches aim to identify effective prompt patterns by analyzing model gradients, reinforcement feedback, or evolutionary processes, reducing the need for manual experimentation.

2.

Tree-of-thought Tree-of-thought prompting generalizes chain-of-thought by generating multiple lines of reasoning in parallel, with the ability to backtrack or explore other paths. It can use tree search algorithms like breadth-first, depth-first, or beam.

3.

Context engineering Context engineering is a related process that focuses on the context elements that accompany user prompts, which include system instructions, retrieved knowledge, tool definitions, conversation summaries, and task metadata. Context engineering is performed to improve reliability, provenance and token efficiency in production LLM systems. The concept emphasises operational practices such as token budgeting, provenance tags, versioning of context artifacts, observability (logging which context was supplied), and context regression tests to ensure that changes to supplied context do not silently alter system behaviour.

4.

MIPRO (Multi-prompt Instruction Proposal Optimizer) optimizes the instructions and few-shot demonstrations of multi-stage language model programs, proposing small changes to module prompts and retaining those that improve a downstream performance metric without access to module-level labels or gradients. **GEPA** (Genetic-Pareto) is a reflective prompt optimizer for compound AI systems that combines language-model-based analysis of execution traces and textual feedback with a Pareto-based evolutionary search over a population of candidate systems; across four tasks, GEPA reports average gains of about 10% over reinforcement-learning-based Group Relative Policy Optimization (GRPO) and over 10% over the MIPROv2 prompt optimizer, while using up to 35 times fewer rollouts than GRPO. Open-source frameworks such as DSPy and Opik expose these and related optimizers, allowing prompt search to be expressed as part of a programmatic pipeline rather than through manual trial and error.

5.

Prompt A prompt is some natural language text that describes and prescribes the task that an artificial intelligence (AI) should perform. A prompt for a text-to-text language model can be a query, a command, or a longer statement referencing context, instructions, and conversation history. The process of prompt engineering may involve designing clear queries, refining wording, providing relevant context, specifying the style of output, and assigning a character for the AI to mimic in order to guide the model toward more accurate, useful, and consistent responses. When communicating with

a text-to-image or a text-to-audio model, a typical prompt contains a description of a desired output such as "a high-quality photo of an astronaut riding a horse" or "Lo-fi slow BPM electro chill with organic samples". Prompt engineering may be applied to text-to-image models to achieve a desired subject, style, layout, lighting, and aesthetic.

6.

Using language models to generate prompts LLMs themselves can be used to compose prompts for LLMs. The automatic prompt engineer algorithm uses one LLM to beam search over prompts for another LLM:

7.

Prompting to estimate model sensitivity Research consistently demonstrates that LLMs are highly sensitive to subtle variations in prompt formatting, structure, and linguistic properties. Some studies have shown up to 76 accuracy points across formatting changes in few-shot settings. Linguistic features significantly influence prompt effectiveness—such as morphology, syntax, and lexico-semantic changes—which meaningfully enhance task performance across a variety of tasks. Clausal syntax, for example, improves consistency and reduces uncertainty in knowledge retrieval. This sensitivity persists even with larger model sizes, additional few-shot examples, or instruction tuning. To address sensitivity of models and make them more robust, several evaluative methods have been proposed. FormatSpread facilitates systematic analysis by evaluating a range of plausible prompt formats, offering a more comprehensive performance interval. Similarly, PromptEval estimates performance distributions across diverse prompts, enabling robust metrics such as performance quantiles and accurate evaluations under constrained budgets.

8.

Prompt injection is a cybersecurity exploit in which adversaries craft inputs that appear legitimate but are designed to cause unintended behavior in machine learning models, particularly large language models. This attack takes advantage of the model's inability to distinguish between developer-defined prompts and user inputs, allowing adversaries to bypass safeguards and influence model behaviour. While LLMs are designed to follow trusted instructions, they can be manipulated into carrying out unintended responses through carefully crafted inputs.

9.

Using gradient descent to search for prompts In "prefix-tuning", "prompt tuning", or "soft prompting", floating-point vectors are searched directly by gradient descent to maximize the log-likelihood on outputs. An earlier result uses the same idea of gradient descent search, but is designed for masked language models like BERT, and searches only over token sequences, rather than numerical vectors. Formally, it searches for $\arg \max_{\tilde{X}} \sum_i \log \Pr[Y^i | X \sim * \tilde{X}^i]$ where $X \sim$ ranges over token sequences of a specified length.

10.

GraphRAG (coined by Microsoft Research) is a technique that extends RAG with the use of a knowledge graph to allow the model to connect disparate pieces of information, synthesize insights,

and understand summarized semantic concepts over large data collections. It was shown to be effective on datasets like the Violent Incident Information from News Articles.

11.

In 2022, text-to-image models like DALL-E 2, Stable Diffusion, and Midjourney were released to the public. These models take text prompts as input and use them to generate images. Early text-to-image models typically do not understand negation, grammar and sentence structure in the same way as large language models, and may thus require a different set of prompting techniques. The prompt "a party with no cake" may produce an image including a cake.

12.

A text-to-image prompt commonly includes a description of the subject of the art, the desired medium (such as digital painting or photography), style (such as hyperrealistic or pop-art), lighting (such as rim lighting or crepuscular rays), color, and texture. Word order also affects the output of a text-to-image prompt. Words closer to the start of a prompt may be emphasized more heavily.

13.

In-context learning A model's ability to temporarily learn from prompts is known as in-context learning. In-context learning is an emergent ability of large language models. It is an emergent property of model scale, meaning that breaks in scaling laws occur, leading to its efficacy increasing at a different rate in larger models than in smaller models. Unlike training and fine-tuning, which produce lasting changes, in-context learning is temporary. Training models to perform in-context learning can be viewed as a form of meta-learning, or "learning to learn".